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Subject: Laboratory Practice VI (410256)
Natural Language Processing Laboratory
Course: - 2019

Assignment No. 03

Name of the Student: _____ **Roll no:** _____

CLASS: - B.E. [Computer]

Division: B/C

Marks: ____/10

Date of Checking: ____/____/____

Sign with Date: _____

ASSIGNMENT NO: 03**TITLE:**

To study basics of text processing using python.

AIM:

Perform text cleaning, perform lemmatization (any method), remove stop words (any method), label encoding. Create representations using TF-IDF. Save outputs.

Theory:**Basics of Data Processing:**

Data processing in natural language processing (NLP) involves transforming raw text into a format suitable for analysis. Below are the basics:

1. Cleaning
2. PoS tagging
3. Stop word removal

**1. Cleaning**

Text cleaning involves removing irrelevant or noisy elements from raw text data to ensure quality.

Common steps include:

- Lowercasing: Convert text to lowercase (e.g., "Hello World" → "hello world").
- Removing special characters: Strip out punctuation, symbols, and emojis.

- Removing numbers: If not relevant, numbers can be removed.
- Whitespace normalization: Replace multiple spaces with a single space.
- URL/email removal: Delete links and email addresses.

Python code:

```
punctuations = "'!()-{}[];:'\"<>./?@$%^&*~`'"
my_str = "Hello!!!, he said ...and went."
no_punct = ""
for char in my_str:
    if(char not in punctuations):
        no_punct = no_punct + char
print(no_punct)
```

Regex expression:

```
import re
s = "Hello!!!, he said ...and went."
s = re.sub(r'^[^\w\s]','',s)
# not of word and space character
print(s)
```

2. Part-of-Speech (PoS) Tagging

PoS tagging involves assigning grammatical labels (e.g., noun, verb, adjective) to each token.

Example:

- Sentence: "The dog runs fast."
- Tags: ["The/DT", "dog/NN", "runs/VBZ", "fast/RB"].
- Applications: Syntax parsing, semantic analysis.
- Tools: NLTK, spaCy, Stanford NLP.

Python code:

```
import nltk
nltk.download('averaged_perceptron_tagger')
from nltk import pos_tag
text = "she sells seashells on the seashore".split()
print("After Split:",text)
tokens_tag = pos_tag(text)
print("After Token:",tokens_tag)
```

3. Stop Word Removal

Stop words are common words (e.g., "is", "the", "and") that may not contribute significant meaning to analysis.

- Removing stop words reduces noise and helps focus on meaningful words.
- Lists of stop words can vary depending on the use case.
- Tools: NLTK, spaCy.

Python code:

```
from nltk.corpus import stopwords  
from nltk.tokenize import word_tokenize  
sentence = "This is a sample sentence, showing off the stop words  
filtration."  
stop_words = set(stopwords.words('english')) # Set of English stop words  
word_tokens = word_tokenize(sentence)      # Tokenize the sentence into  
words  
# Filtering the sentence  
filtered_sentence = [w for w in word_tokens if not w.lower() in stop_words]  
print("Original tokens:", word_tokens)  
print("Filtered tokens:", filtered_sentence)
```

Conclusion:

These steps form the foundation of preprocessing, which is crucial for downstream tasks such as sentiment analysis, machine learning models, and information retrieval.

Questions:

- Q1. Explain Components of Natural Language Processing in details.
- Q2. Explain concept of Finite automata for NLP with example.
- Q3. Explain Applications of Natural Language Processing- Chatbot in details.